

Question	Answer	Marks
3(a)	total kinetic energy associated with random motion of molecules	B1
	potential energy (of molecules) is zero	B1
3(b)(i)	$pV = NkT$ and $U = (3/2)NkT$	C1
	$U = (3/2)pV$ $= (3/2) \times 2.0 \times 10^5 \times 0.016$ $= 4800 \text{ J}$	A1
3(b)(ii)	temperature = 800 K	A1
3(b)(iii)	volume = 0.032 m ³	A1
3(b)(iv)	straight vertical line labelled X between A and (0.016, 4.0)	B1
	curve from B with continuously decreasing negative gradient, labelled Y	B1
	line labelled Y between (0.016, 4.0) and (0.032, 2.0)	B1

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